

PROJECT HOOGLAND

Commercial Performance Diagnostic

Project Delivery & Architecture Report

Data Engineering · Advanced Analytics · Business Intelligence

Prepared by: Harshil Patel

Programme: MBA Data Analytics,
Wittenborg

Date: May 2026

Classification: Confidential

1. Executive Summary & Business Case

1.1 The Problem

Hoogland Outdoor operates 55 stores across the Netherlands with two years of transactional data totalling €379M in net revenue. Despite a healthy topline, the business has no systematic visibility into where margin is being destroyed, which assets are underperforming, or whether its commercial programmes (promotions, loyalty, channel strategy) are creating or eroding value. Management makes investment decisions based on revenue growth alone — a metric that masks the margin economics underneath.

Without a diagnostic capability, the business faces three structural risks. First, a promotional calendar that runs continuously without margin accountability. Second, a product catalogue that ties up capital in slow-moving inventory. Third, a store estate with significant performance variance that is neither measured nor managed.

1.2 The Approach

Project Hoogland was designed as a four-week diagnostic sprint, structured in two phases. Phase 1 (Weeks 1–2) built the data infrastructure: a modern analytics warehouse using dbt and DuckDB, transforming raw transactional CSVs through three layers (staging, intermediate, mart) into five business-ready analytical tables. Phase 2 (Weeks 3–4) built two consumer layers on top: a six-page Power BI dashboard for management reporting, and a Python-based Jupyter Notebook for hypothesis-driven statistical analysis.

The entire stack was designed for auditability. Every number on every dashboard page traces back to a single mart. Every mart traces back through documented dbt transformations to the raw source. There is one source of truth, and two lenses applied to it.

1.3 The “So-What”: Key Findings

Three numbers for the board

1) €40.5M in margin destroyed by the promotional calendar over two years (€20.3M/year). Every single campaign runs below the 45.4% baseline margin. Uitverkoop and Black Friday alone account for €14M.2) €116M in working capital trapped in long-tail SKUs. The bottom 50% of the 620-SKU catalogue generates just 4.6% of unit sales. Apparel Technical alone holds €50M in dead stock.3) €7.1M annual revenue gap from 25 stores underperforming their archetype peers. City centre stores at Amersfoort, Utrecht, and Arnhem have the largest gaps.

#	Hypothesis	Verdict	€ Impact	Conf.	Priority
H1	Promo ROI illusion	Confirmed	€40.5M	High	1
H6	Pareto SKU concentration	Confirmed	€116M	High	2
H2	Store allocation variance	Confirmed	€7.1M	Medium	3
H5	Loyalty asymmetry	Confirmed	€4.3M	Medium	4
H4	Weather-demand coupling	Confirmed	€1.7M	High	5

H3	Channel cannibalization	Rejected	No risk	High	—
----	-------------------------	----------	---------	------	---

1.4 Business Value & Recommended Actions

The diagnostic identifies a total addressable margin uplift of approximately €53.6M and €116M in trapped working capital eligible for release. Three priority actions for the next 90 days:

- Restructure the promotional calendar: Kill or redesign uitverkoop and Black Friday campaigns. Shift to conditional mechanics (buy-2-get-1, loyalty exclusives, bundles). Protect gifting campaigns. Target: €8M/year in margin recovery.
- Launch range rationalisation in Apparel Technical: Exit or consolidate SKUs with fewer than 500 annual units and more than €300K carrying cost. A 30% tail reduction releases approximately €35M in working capital.
- Diagnose the bottom 10 stores: Conduct a store-by-store review of Amersfoort, Utrecht, Arnhem, and the seven other underperformers. Test staffing, layout, competition, and catchment drivers. Close the €7.1M gap.

2. Strategic Technical Approach

2.1 Design Philosophy

The technology choices for Project Hoogland were not driven by novelty or preference. They were driven by three non-negotiable requirements: auditability (every number must trace to its source), reproducibility (any analyst can re-run the entire pipeline and get identical results), and speed (the full diagnostic must complete within four weeks, with one analyst, on local hardware).

These constraints eliminated several common approaches. A cloud data warehouse (Snowflake, BigQuery) would introduce unnecessary cost and provisioning overhead for a 165MB dataset. A traditional RDBMS (PostgreSQL, MySQL) is optimised for transactional workloads, not the columnar scans required for analytical aggregation across millions of rows. Manual Excel analysis would be neither auditable nor reproducible.

2.2 Why DuckDB

DuckDB is an in-process analytical database engine. It runs locally inside the Python process or the dbt project — no server, no network, no authentication. It is purpose-built for the workload this project demands: scanning and aggregating large tables with complex filters and groupings.

- **Columnar storage:** DuckDB stores data in columns rather than rows, which means an aggregation across `net_revenue_eur` scans only that column, not the entire table. For the 2.3M-row `fact_sales` table, this is an order-of-magnitude performance difference versus row-oriented databases.
- **Zero infrastructure:** The entire warehouse is a single 165MB file (`hoogland.duckdb`). It can be copied, backed up, version-controlled, and shared without any server configuration.
- **Dual-consumer compatibility:** Both Power BI (via the MotherDuck .mez connector) and Python (via the `duckdb` package) read from the same file. One source of truth, two consumers, zero reconciliation drift.

2.3 Why Python and Jupyter Notebook for EDA

The EDA notebook serves a fundamentally different purpose than the Power BI dashboard. The dashboard answers what is happening. The notebook answers why it is happening, whether the pattern is statistically real, and how much it costs in euros.

This requires capabilities that Power BI does not provide: linear regression (`scipy.stats.linregress`), correlation testing (Pearson r with p -values), decile analysis, and the ability to programmatically sweep across all nine product categories or all 55 stores in a single loop. Python with `pandas`, `scipy`, and `matplotlib` is the standard toolkit for this class of work across consulting, finance, and academic research.

The notebook was structured as a consulting deliverable, not an exploratory scratchpad. Each hypothesis section follows a fixed template: claim stated, method defined, data pulled, statistical test executed, finding quantified in euros, recommendation written. The notebook is designed to be read by a non-technical stakeholder, with every line of code commented and every output interpreted in business language.

2.4 Technology Stack Summary

Layer	Tool	Strategic Rationale
Storage	DuckDB	Columnar, serverless, single-file. Eliminates infrastructure cost and reconciliation risk.

Transformation	dbt Core	SQL-based, version-controlled, three-layer lineage. Auditability and reproducibility by design.
BI Layer	Power BI Desktop	Industry-standard dashboard tool. Connected via MotherDuck .mez connector. One mart per page.
Analytics	Python 3.11 + Jupyter	Statistical hypothesis testing, regression, correlation. Consulting-grade reproducible analysis.
Libraries	pandas, scipy, statsmodels, matplotlib, seaborn	DataFrame manipulation, inferential statistics, and publication-quality visualisation.

3. Data Engineering & Analytics Engineering Framework

3.1 Why dbt

dbt (data build tool) is the transformation layer of the modern analytics stack. It takes raw data already loaded into a warehouse and transforms it through modular, tested, documented SQL models. In Project Hoogland, dbt replaced what would otherwise be a series of ad hoc SQL scripts or Python transformations with a governed, version-controlled pipeline.

The choice of dbt was driven by three principles:

- **Modularity:** Each transformation is a single SQL file with a clear input and output. A change to the staging layer does not require rewriting the marts.
- **Lineage:** dbt automatically tracks which models depend on which. The command `dbt docs generate` produces a full dependency graph showing exactly how raw CSVs flow through to the five analytical marts.
- **Testing:** dbt natively supports schema tests (`not_null`, `unique`, `accepted_values`, `relationships`) that run on every build. All 23 models pass validation.

3.2 The Three-Layer Architecture

The dbt project (`dbt_hoogland`) implements the industry-standard three-layer transformation pattern. Each layer has a distinct role, a clear naming convention, and a defined contract with the layers above and below it.

Layer 1: Staging (`stg_*`)

Thirteen staging models clean the raw CSV source data. Staging models rename columns to a consistent convention, cast data types (strings to dates, integers to decimals), drop malformed or duplicate rows, and apply no business logic. The staging layer is a contract: everything downstream can trust that dates are dates, IDs are strings, and nulls have been handled.

Model	Source & Purpose
<code>stg_fact_sales</code>	Transaction-level sales: 2.3M rows, grain = line item per transaction
<code>stg_fact_inventory_weekly</code>	Weekly stock positions: opening/closing units, receipts, by store and SKU
<code>stg_fact_returns</code>	Return transactions linked to original sales
<code>stg_fact_supply_orders</code>	Purchase orders to suppliers
<code>stg_fact_traffic_daily</code>	Daily footfall by store
<code>stg_dim_store</code>	55 stores with city, region, archetype, sqm, channel
<code>stg_dim_product</code>	620 SKUs with category, subcategory, seasonality, lifecycle
<code>stg_dim_customer</code>	80K customers with segment, tier, demographics
<code>stg_dim_brand</code>	Brand reference data
<code>stg_dim_campaign</code>	18 campaigns with type, start/end dates
<code>stg_dim_date</code>	Calendar dimension
<code>stg_ext_weather_daily</code>	Daily weather: temperature, precipitation, wind by city

stg_campaign_sku_mapping	Many-to-many mapping of campaigns to SKUs
---------------------------------	---

Layer 2: Intermediate (int_*)

Five intermediate models join and enrich the staged data. This is where business logic enters the pipeline: sales records are joined to store attributes, products are enriched with brand and campaign metadata, and weather data is attached to the correct store-city-date combination. Intermediate models are not exposed to consumers. They exist solely to make the mart logic readable and maintainable.

Model	Join Logic
int_sales_enriched	fact_sales + dim_store + dim_product + dim_brand + dim_customer
int_sales_with_campaign	int_sales_enriched + dim_campaign + campaign_sku_mapping
int_customer_metrics	Customer-level aggregation: RFM scores, CLV, tenure, recency
int_inventory_enriched	fact_inventory_weekly + dim_store + dim_product: stock health metrics
int_weather_store_daily	ext_weather_daily + dim_store: weather attached to store geography

Layer 3: Marts (mart_*)

Five mart models are the final, consumer-facing analytical tables. Each mart is designed to answer a specific set of business questions and is consumed by exactly one Power BI dashboard page and one section of the EDA notebook. The one-mart-per-page constraint is an architectural decision, not a limitation: it prevents join ambiguity and ensures every number is auditable.

Mart	Grain	Key Metrics	Consumers
mart_store_performance	Store × Day	Revenue, margin, basket, rev/sqm	P1, P2, H2, H3
mart_promo_effectiveness	Campaign	Revenue, margin, discount depth	P3, H1
mart_customer_clv	Customer	CLV, RFM, tenure, recency	P4, H5
mart_inventory_health	Store × SKU × Week	Stock, WoC, lost revenue	P5, H6
mart_weather_demand	Store × Category × Day	Revenue, temp, precip, sensitivity	P6, H4

3.3 Data Quality & Reconciliation

A reconciliation gate was enforced at every entry point. Before any dashboard page was built or any hypothesis tested, the mart totals were compared against the locked baseline: gross €393M, net €379M, margin 45.35%, basket €162, 55 stores. The notebook reconciliation cell confirmed: gross €393.1M, net €379.3M, margin 45.35%, basket €162.06, 55 stores — all within rounding tolerance. This gate ensures that every finding in the EDA and every KPI on the dashboard ties back to the same verified source.

4. dbt Architecture & Data Lineage

4.1 Project Structure

The dbt project is organised in the following directory structure, following dbt best practices for modularity and discoverability:

Directory	Contents
models/staging/	13 stg_* models: one per source CSV. Clean, cast, rename. No business logic.
models/intermediate/	5 int_* models: join, enrich, compute. Business logic layer.
models/marts/	5 mart_* models: consumer-ready analytical tables. Final schema.
seeds/	Raw CSV source files loaded via dbt seed.
tests/	Schema tests (not_null, unique, relationships) applied across all layers.
macros/	Reusable SQL macros for common transformations.

4.2 Materialisation Strategy

All models are materialised as tables (not views) in DuckDB. This decision trades storage space for query performance: since both Power BI and the Python notebook perform full-table scans and aggregations, pre-materialised tables eliminate redundant computation. The full pipeline runs in under 30 seconds on local hardware, making the storage-performance trade-off unambiguously positive.

4.3 Data Lineage

The complete lineage from source to consumer follows this path:

Lineage Flow

Raw CSVs (seeds/) → Staging (stg_*): clean, cast, rename → Intermediate (int_*): join, enrich, compute → Marts (mart_*): aggregate to business grain → Consumers: Power BI (6 pages) + Python EDA (6 hypotheses)

Each mart draws from one or two intermediate models, which in turn draw from multiple staging models. The dependency graph is a clean DAG (directed acyclic graph) with no circular references and no cross-layer shortcuts. A change to any staging model automatically propagates through the correct intermediate and mart models on the next dbt run.

4.4 Key Schema Decisions

- Store Performance mart grain is store × date (daily). This enables time-series analysis in both Power BI (monthly trend) and Python (daily regression). A weekly grain was considered but rejected because weather analysis requires daily resolution.
- Promo Effectiveness mart grain is campaign (one row per campaign). This was a deliberate simplification: a SKU × store × week grain would enable full Difference-in-Differences but would produce a table too large for Power BI's in-memory model. The campaign-level grain supports the margin-gap analysis that delivered the €40.5M finding.

- Customer CLV mart computes RFM scores (recency, frequency, monetary) as integer scores 1–5 within the dbt model, not in Python or Power BI. This ensures the scores are computed once, consistently, and are available to both consumers.
- Inventory Health mart includes estimated lost revenue from stockouts, computed in the intermediate layer by multiplying the average daily sales rate by the number of stockout days. This is a conservative estimate and is flagged as such in the dashboard.
- Weather Demand mart includes a `weather_sensitivity` score and a `monthly_temp_anomaly` field, both computed in the intermediate layer. The sensitivity score is used for category-level regression in the Python notebook.

5. Advanced Analytics: EDA & Hypothesis Testing

5.1 Methodology

Six business hypotheses were defined at project inception, each targeting a specific commercial risk or opportunity. The hypotheses were tested in order of analytical complexity (easiest to hardest), not numerically. This sequencing builds confidence with the data and tools before tackling complex statistical methods, and delivers quick wins early.

Each hypothesis was evaluated using a consistent framework:

- **Claim:** A falsifiable business statement (e.g., “Promotions destroy margin”).
- **Method:** The specific statistical technique and data source used to test the claim.
- **Evidence:** The quantitative result, including effect size, significance level, and confidence.
- **Euro impact:** The business cost or opportunity, expressed in euros, annualised where appropriate.
- **Recommendation:** A specific, actionable directive for management.

5.2 H6 — Pareto: SKU Revenue Concentration

Claim: A small percentage of SKUs generates most revenue, and the long tail costs more than it earns.

Method: Ranked all 620 SKUs by total units sold from mart_inventory_health. Computed cumulative unit share and cross-referenced bottom-50% inventory carrying costs.

Result: Confirmed — sharper than classic 80/20

Top 20% of SKUs generate 81.8% of all units sold. Bottom 50% (310 SKUs) contribute 4.6% of units but hold €116M in inventory. Apparel Technical: 118 SKUs, €50M trapped (43% of dead stock). Hiking: 62 SKUs, €23M. Ski & Winter: 64 SKUs, €21M.

Recommendation: Range rationalisation starting with Apparel Technical. A 30% tail reduction releases approximately €35M in working capital.

5.3 H2 — Store Allocation Variance

Claim: Some stores are systematically under-resourced relative to their archetype peers.

Method: Computed revenue per square metre across 54 physical stores (excluding the ecommerce DC). Segmented by archetype and calculated the gap between each store and its archetype median.

Result: Confirmed — 25 of 54 stores below peer median

Revenue density varies 5.5x across archetypes: city centre €24/sqm vs outlet €4/sqm. Coefficient of variation: 29.7%. The three largest gaps: Amersfoort (€-880K/year), Utrecht (€-818K/year), Arnhem (€-787K/year). Total annual revenue gap: €7.1M.

Recommendation: Store-by-store diagnostic on the bottom 10. Test staffing, layout, competition, and catchment. Prioritise Amersfoort and Utrecht for highest upside per intervention euro.

5.4 H4 — Weather-Demand Coupling

Claim: Certain product categories are meaningfully sensitive to weather, creating forecastable revenue risk.

Method: Linear regression (scipy.stats.linregress) of daily category revenue against average temperature for each of nine categories. 6,579 observations across 2024–2025.

Category	€/day per °C	R ²	p-value	Direction
Hiking	+€8,089	0.38	<0.001	Warm = more sales
Ski & Winter	-€5,203	0.42	<0.001	Warm = fewer sales
Camping	+€644	0.65	<0.001	Warm = more sales
Apparel Lifestyle	+€138	0.00	0.56	Not significant

Result: Confirmed — 7 of 9 categories significant

A warm season (+5°C, 120 days) shifts +€4.9M to Hiking but -€3.1M from Ski & Winter. Net exposure: €1.7M. Camping has the tightest model fit (R² = 0.65).

Recommendation: Weather-adjusted demand forecasting for inventory planning. Pre-position warm-weather stock when forecasts trend warm. Hedge Ski & Winter with flexible supplier contracts.

5.5 H5 — Loyalty Programme Asymmetry

Claim: The loyalty programme disproportionately benefits the top tier.

Method: Compared CLV, RFM scores, and transaction frequency between the two member tiers (Plus and Plus Premium) across 80,000 customers. Performed a decile analysis of CLV across the full base.

Result: Confirmed — but asymmetry is within the base, not between tiers

The two tiers are statistically identical: avg CLV €2,644 vs €2,659, same RFM scores (4.3/4.6/4.7), same recency (54 days). The tier structure is decorative. The real asymmetry: top decile CLV €7,423 vs bottom decile €433 — a 17.2x gap. Top 10% generate 28.1% of total CLV.

Recommendation: Scrap the two-tier structure. Replace with 3–4 tiers based on RFM scores: Platinum (D9–D10), Gold (D6–D8), Silver (D1–D5). A 5% retention lift in the top decile protects approximately €2.8M in CLV.

5.6 H3 — Channel Cannibalization

Claim: Online growth is eating into physical store revenue.

Method: Pearson correlation between monthly ecommerce and store growth rates. Year-over-year revenue comparison by channel.

Result: Rejected — channels are complementary

Monthly growth correlation: $r = 0.995$ ($p < 0.0001$). Both grew YoY: ecommerce +3.5% (€51.9M → €53.7M), stores +2.8% (€135.0M → €138.8M). Ecommerce share stable at 27.8–27.9%.

Recommendation: Stop treating channels as zero-sum. Invest in ecommerce growth targeting 35% share by FY27 without defensive store cuts. Build cross-channel features: click-and-collect, unified loyalty, in-store returns for online orders.

5.7 H1 — Promo ROI Illusion

Claim: Promotions generate revenue but destroy margin, creating a false ROI.

Method: Compared each of 18 campaigns against the 45.4% baseline margin. Calculated margin destruction as the margin-percentage gap multiplied by gross revenue for each campaign.

Type	Count	Revenue	Avg Margin	Avg Disc.	Destroyed
Campaign	8	€156.4M	35%	17%	-€19.5M
Uitverkoop	4	€23.8M	16%	38%	-€11.5M
Black Friday	2	€19.5M	25%	30%	-€5.5M
Gifting	4	€64.7M	40%	11%	-€4.0M

Result: Confirmed — €40.5M in margin destroyed

Every single campaign runs below baseline margin. Uitverkoop drops to 14–18% margin (30pp below baseline). Gifting is the only format approaching acceptable levels at 39–40%. Total destruction over two years: €40.5M (€20.3M/year).

Recommendation: Kill or restructure uitverkoop and Black Friday (€14M combined destruction). Shift to conditional mechanics. Protect gifting. Target: €8M/year margin recovery.

6. Business Intelligence Delivery

6.1 Dashboard Architecture

The Power BI dashboard (Hoogland_Dashboard.pbix) comprises six pages, each mapped to a single analytical mart via the MotherDuck DuckDB connector. This one-mart-per-page constraint is an architectural decision that ensures auditability: every visual on every page traces to a single SQL source with no join ambiguity.

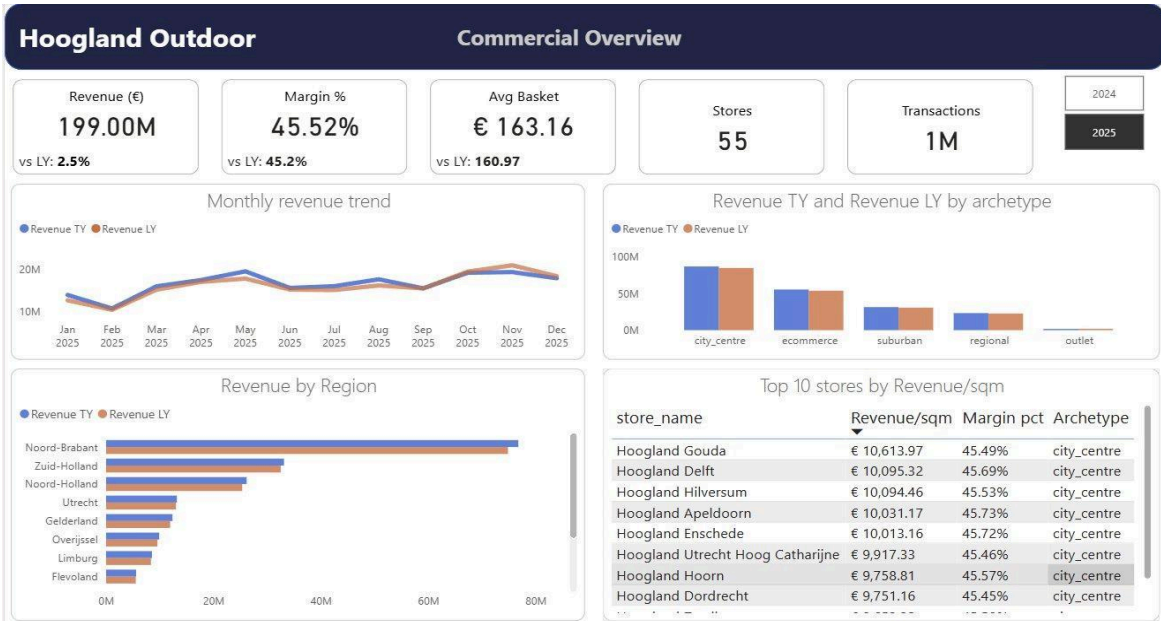
A shared Calendar DAX table provides the date dimension across all pages, with year-over-year measures (TY/LY revenue, margin, basket, transactions, and delta percentage) and a year slicer enabling toggle between 2024, 2025, or combined view.

6.2 Dashboard Matrix

Page	Title	Mart Source	Business Question
P1	Commercial Overview	mart_store_performance	What is the overall commercial health of the business?
P2	Store Performance	mart_store_performance	Which stores over/underperform and why?
P3	Promo Effectiveness	mart_promo_effectiveness	Are promotions creating or destroying value?
P4	Customer CLV	mart_customer_clv	Who are our most valuable customers and does loyalty work?
P5	Inventory Health	mart_inventory_health	Where is capital trapped and where are we losing sales?
P6	Weather Demand	mart_weather_demand	How does weather affect category demand?

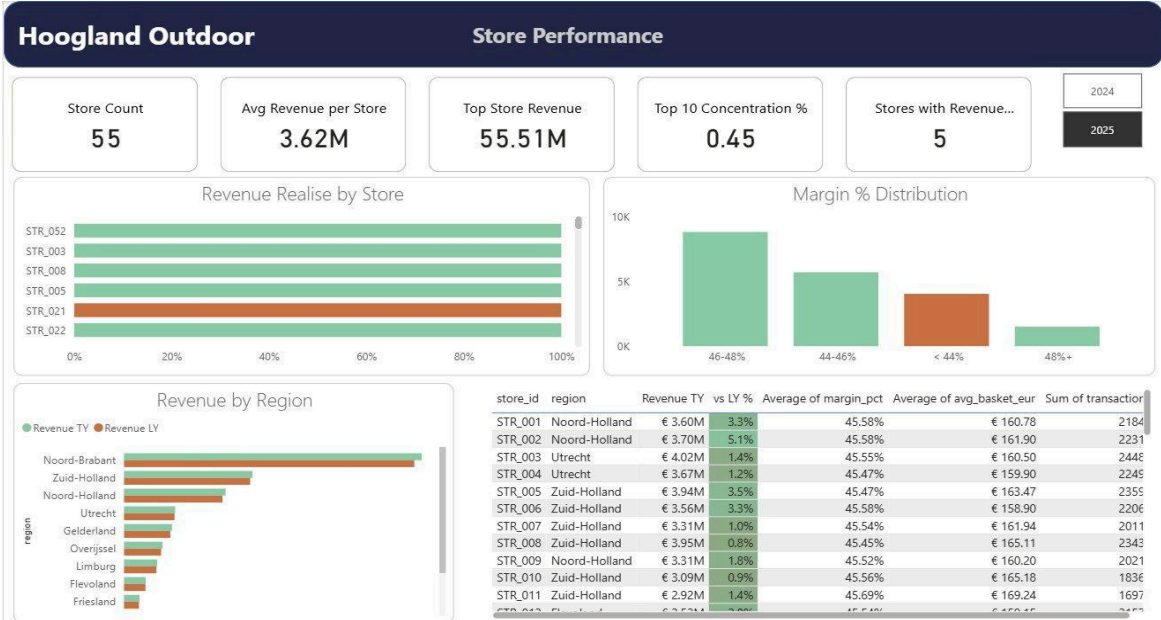
6.3 Page 1 — Commercial Overview

The anchor page provides executive-level KPIs and trend context. Five cards display net revenue (€199M for 2025, +2.5% vs LY), margin (45.52%), average basket (€163.16), store count (55), and transaction volume (1M). A monthly TY/LY trend line, revenue by region bar chart, revenue by archetype comparison, and a top-10 stores table ranked by revenue per square metre complete the view.



6.4 Page 2 — Store Performance

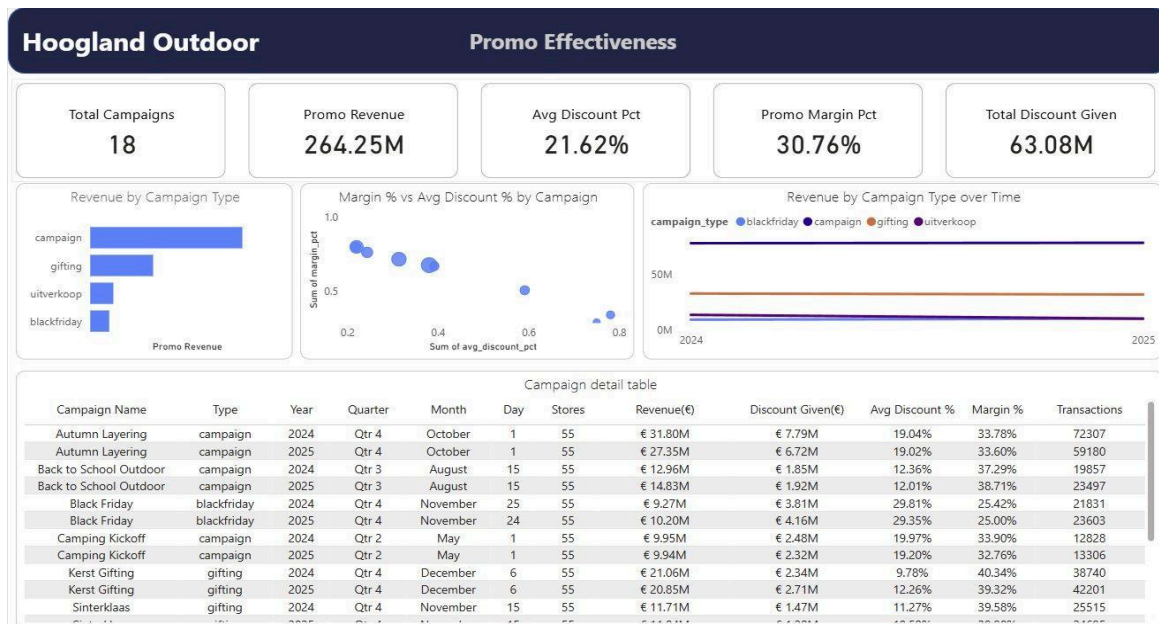
Designed for operational managers to identify outlier stores. Features a revenue-realised-by-store bar chart showing relative performance across all 55 locations, a margin percentage distribution histogram segmented into bands (below 44%, 44–46%, 46–48%, above 48%), revenue by region with TY/LY comparison, and a sortable detail table with store-level metrics including revenue, YoY growth, margin, basket, and transactions.



6.5 Page 3 — Promotional Effectiveness

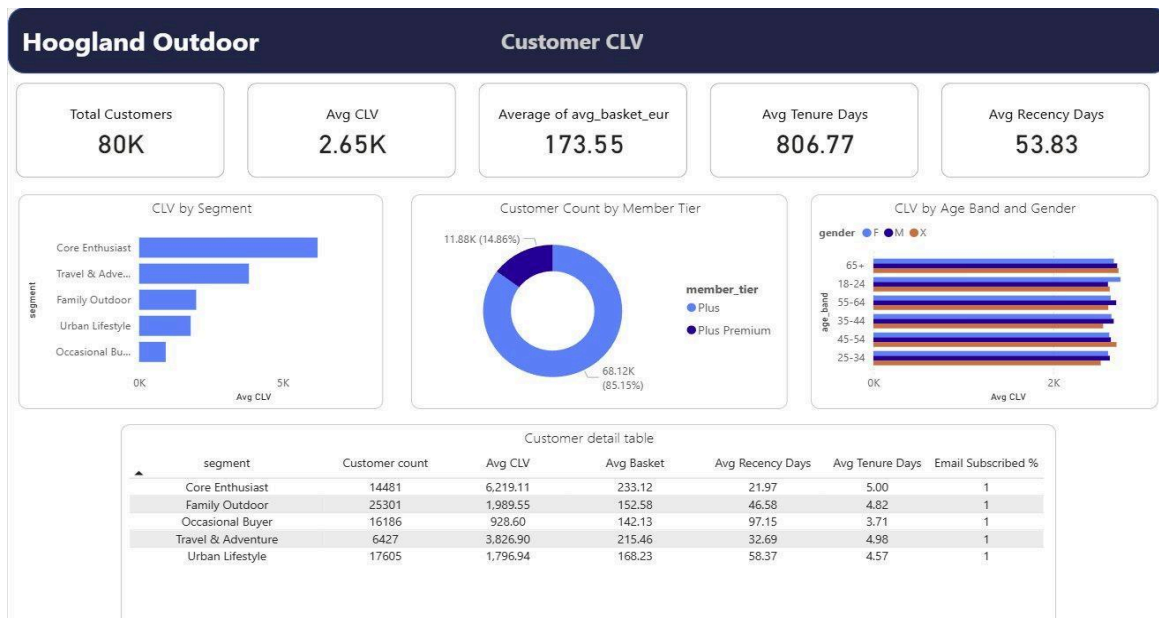
The most commercially consequential page. Displays 18 campaigns across four types with total promo revenue of €264.25M, average discount of 21.62%, and promo margin of 30.76%. A revenue-by-type bar chart, a margin-vs-discount scatter plot (showing the inverse relationship

between discount depth and margin), and a campaign timeline visual expose the promotional calendar structure. The detail table enables campaign-by-campaign inspection.



6.6 Page 4 — Customer CLV

Exposes the loyalty programme structure across 80,000 customers. KPI cards show average CLV (€2.65K), average basket (€173.55), tenure (807 days), and recency (54 days). CLV by segment reveals Core Enthusiasts at €6,219 versus Occasional Buyers at €929. The member tier donut confirms the 85/15 split between Plus and Plus Premium. A CLV-by-age-band-and-gender breakdown and a segment detail table complete the view.



6.7 Page 5 — Inventory Health

Total inventory value of €10.28M across 4M sales units, with 363 stockout SKUs, 4 overstock SKUs, and an average 79.86 weeks of cover. Inventory value by category shows Hiking and Apparel Technical as the largest holders. The lifecycle stage breakdown (core, mature, end-of-life, new) reveals that core products dominate inventory value. A stock status donut confirms 90.27% healthy, 8.7% at risk. The detail table enables drill-through by category and stock status.



6.8 Page 6 — Weather Demand

Connects weather data to commercial performance. KPI cards show average temperature (11.22°C), weather sensitivity (0.32), precipitation (1.96mm), wind speed (13.01 km/h), and temperature anomaly (+0.85°C). Revenue by temperature band confirms that mild and cold conditions drive the highest revenue. Revenue by precipitation band shows dry and light-rain conditions dominate. The detail table enables region-by-category inspection of weather-revenue relationships.



7. Conclusion

Project Hoogland delivered a complete commercial diagnostic in four weeks. The infrastructure layer (dbt + DuckDB) provides a governed, auditable, single-source-of-truth warehouse. The business intelligence layer (Power BI) gives management an interactive view of what is happening. The analytical layer (Python EDA) explains why it is happening and quantifies the cost.

Five of six hypotheses were confirmed. The three highest-priority findings are:

- The promotional calendar destroys €20.3M in margin annually. Every campaign runs below baseline. Restructuring the bottom quartile recovers approximately €8M per year.
- The bottom half of the 620-SKU catalogue generates 4.6% of sales but traps €116M in working capital. Range rationalisation in Apparel Technical alone releases €35M.
- Twenty-five stores underperform their archetype peers by a combined €7.1M annually. A targeted diagnostic on the bottom 10 stores has the highest ROI.

One hypothesis was conclusively rejected: there is no evidence of channel cannibalization. Ecommerce and stores move in near-perfect lockstep ($r = 0.995$). The strategic risk is under-investment in ecommerce, not over-investment.

Total addressable value

Margin uplift: €53.6M addressable across all confirmed findings. Working capital release: €116M in trapped long-tail inventory. The data infrastructure built in this engagement is reusable: any future hypothesis can be tested against the same warehouse, extending the diagnostic capability beyond this initial sprint.